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- Basic two-body equation of relative motion:

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where $\mu = GM$,
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- For earth, $\mu_E := 3.986004418 \times 10^{14}$ m^3/sec^2 (from WGS84)

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- Energy is also conserved (if we ignore internal, heat-generating forces)

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  - Corrects Newtonian formula given in Eq. (1)

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- Other minor effects: thermal, magnetic, ...

Linearization about a Nominal Trajectory: Taylor Series

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where $D^k_{x,x_0} f := \left(\sum_{i=1}^{n} (x_i - [x_0]_i) \frac{\partial f}{\partial x_i}\Big|_{x_0} \right)^k$, and it is understood that the exponent k applies to the derivative operator to form the kth-order derivative **before** evaluating that derivative at the vector point x_0.

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- Note that x is also (in our applications) an implicit vector-valued function of time $x = x(t)$. But we are ignoring that for now.

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$$D^k_{x,x_0} f := \begin{cases} (x - x_0)^k \dfrac{d^k f}{dx^k}\Big|_{x_0}, & \text{if } \dim(x) = 1; \\ \left(\displaystyle\sum_{i=1}^{n} (x_i - [x_0]_i) \frac{\partial f}{\partial x_i}\Big|_{x_0} \right)^k, & \text{if } \dim(x) > 1. \end{cases} \tag{4}$$

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- $x(t) = F\left[t, t_0, x_0(t)\right]$
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- $\dot{x}(t) = \dot{x}_{\text{ref}}(t) + \left[\dfrac{\partial F(t)}{\partial \lambda}\right]_{\lambda = x_{\text{ref}}} \left[x(t) - x_{\text{ref}}(t)\right] + \cdots,$
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- Or, equivalently
  $\dot{x}(t) = F(x^*, t) + \left[\dfrac{\partial F(x,t)}{\partial x(t)}\right]^* \left[x(t) - x^*(t)\right] + \cdots$

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- Note: Equilibrium points are called critical points in mathematics (points where the gradient is zero)

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- Then the equation of motion becomes: $\dot{x}(t) = Ax(t)$, where $A(t) := \left[\frac{\partial \dot{X}(t)}{\partial X(t)}\right]^*$
- The state transition matrix Φ is associated with the homgeneous solution of the linearized differential equation above, and enables future state to be computed from the current state :

$$x(t) = \Phi(t, t_0)x(t_0)$$

and also itself satisfies the matrix differential equation

$$\dot{\Phi}(t, t_0) = A(t, t_0)\Phi(t, t_0),$$

with initial condition $\Phi(t_0, t_0) = I$

Simplified Perturbation Models

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- SGP: Simplified General Perturbations (orbital period ¡ 225 minutes)
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- Many implementations available: Python, MATLAB, FORTRAN, C

Example: 2D, 7-state

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- Consider a simple, planar example: $X = \left[X_1, X_2, \dot{X}_1, \dot{X}_2, \mu, X_s, Y_s\right]^\top$, where X_1, X_2 are the two SV position coordinates and the γ (nuisance) parameters are μ and the monitor station coordinates X_s, Y_s.

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- Note that $\dot{X} = \left[\dot{X}_1, \dot{X}_2, \ddot{X}_1, \ddot{X}_2, \dot{\mu}, \dot{X}_s, \dot{Y}_s\right]^\top$

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- Note that $\dot{X} = \left[\dot{X}_1, \dot{X}_2, \ddot{X}_1, \ddot{X}_2, \dot{\mu}, \dot{X}_s, \dot{Y}_s\right]^\top$
- Using $\ddot{X}_1 = -\mu X_1/r^3$ and $\ddot{X}_2 = -\mu X_2/r^3$, and assuming that μ and the station coordinates are constant, we obtain $\dot{X} = \left[\dot{X}_1, \dot{X}_2, -\frac{\mu X_1}{r^3}, -\frac{\mu X_2}{r^3}, 0, 0, 0\right]^\top$

Example: 2D, 7-state Model

Measurement Model

Example: 2D, 7-state Model

Measurement Model

- The A matrix becomes:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ -\frac{\mu}{r^3} + \frac{3\mu X_1^2}{r^5} & \frac{3\mu X_1 X_2}{r^5} & 0 & 0 & -\frac{X_1}{r^3} & 0 & 0 \\ \frac{3\mu X_1 X_2}{r^5} & -\frac{\mu}{r^3} + \frac{3\mu X_2^2}{r^5} & 0 & 0 & -\frac{X_2}{r^3} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^*$$

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- If the A matrix were constant, we could easily calculate the state transition matrix Φ to be the matrix exponential of A:

$$\Phi(t - t_0) = \exp\left[A((t - t_0)\right]$$

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- We evaluate A at X_0
- We have to integrate the Φ differential equation from t_0 to t get the new value of the state transition matrix that will allow us to extrapolate the state from time t_0 to t, keeping it mind that X_1, X_2, r are functions of time.

Example: 2D 7-state Model

System Dynamics Matrix

$$A(t) = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{\mu\left(2X^2-Y^2\right)}{(X^2+Y^2)^{5/2}} & \frac{3XY\mu}{(X^2+Y^2)^{5/2}} & 0 & 0 & -\frac{X}{(X^2+Y^2)^{3/2}} & 0 & 0 \\ \frac{3XY\mu}{(X^2+Y^2)^{5/2}} & -\frac{\mu\left(X^2-2Y^2\right)}{(X^2+Y^2)^{5/2}} & 0 & 0 & -\frac{Y}{(X^2+Y^2)^{3/2}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

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- We linearize this by taking the Jacobian of this to get the time-dependent H matrix:

$$H := \frac{\partial \rho}{\partial X} = \left[\frac{X - X_s}{\rho} \quad \frac{Y - Y_s}{\rho} \quad 0 \quad 0 \quad 0 \quad -\frac{X - X_s}{\rho} \quad -\frac{Y - Y_s}{\rho} \right]$$

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- If measurements arrive at different times after epoch, then they can be related back to epoch state by applying Φ: Let $H(i)$ be the H matrix at time $t_i > t_k$. Define new observation matrix Γ whose i^{th} element is the row block $H(i)\Phi(t_i, t_k)$

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- Stack all measurements via $z = \Gamma x(k) + v(k)$

GPS OCX

Orbit Integrator (OI)

- Equation of motion: $\ddot{r}(t) = f(r, \dot{r}, p, t)$, where p is a vector of constant force model parameters

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- Force models:

  $f = f_{grav} + f_{3b} + f_{srp} + f_{tides} + f_{rel} + f_{earthrad} + f_{custom}$ where
  f_{grav} = acceleration due to gravity
  f_{2b} = acceleration due to sun and/or moon f_{srp} = acceleration due to solar radiation pressure
  f_{tides} = acceleration due to to solid earth, pole, and ocean tides
  f_{rel} = acceleration due to relativity
  $f_{earthrad}$ = acceleration due to earth albedo and IR radiation
  f_{custom} = acceleration due to custom force model (e.g., constant and once-per-rev accelerations)

  See gcore.pdf for details on the force models

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- Variational Equations:

$$\frac{\partial^2}{\partial t^2}\left(\frac{\partial r}{\partial r_0}\right) = A\frac{\partial r}{\partial r_0} + B\frac{\partial \dot{r}}{\partial r_0} \tag{5}$$

$$\frac{\partial^2}{\partial t^2}\left(\frac{\partial r}{\partial \dot{r}_0}\right) = A\frac{\partial r}{\partial \dot{r}_0} + B\frac{\partial \dot{r}}{\partial \dot{r}_0} \tag{6}$$

$$\frac{\partial^2}{\partial t^2}\left(\frac{\partial r}{\partial p}\right) = A\frac{\partial r}{\partial p} + B\frac{\partial \dot{r}}{\partial p} + C \tag{7}$$

- To facilitate filter estimation of the states, we compute partial derviatives of the state with respect to $r_0, \dot{r}_0$ and p; partials represent sensitivity of satellite state to uncertainty in initial conditions (i.e., prior state)
- Variational Equations:

$$\frac{\partial^2}{\partial t^2}\left(\frac{\partial r}{\partial r_0}\right) = A\frac{\partial r}{\partial r_0} + B\frac{\partial \dot{r}}{\partial r_0} \tag{5}$$

$$\frac{\partial^2}{\partial t^2}\left(\frac{\partial r}{\partial \dot{r}_0}\right) = A\frac{\partial r}{\partial \dot{r}_0} + B\frac{\partial \dot{r}}{\partial \dot{r}_0} \tag{6}$$

$$\frac{\partial^2}{\partial t^2}\left(\frac{\partial r}{\partial p}\right) = A\frac{\partial r}{\partial p} + B\frac{\partial \dot{r}}{\partial p} + C \tag{7}$$

where $A = \frac{\partial f}{\partial r}, B = \frac{\partial f}{\partial \dot{r}}$ and $C = \frac{\partial f}{\partial p}$

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Orbit Integrator (OI)

- Variational equations must be integrated together with equations of motion; A, B, and C need to be computed at every time step

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- Force models internally hard-coded as modular suboutines; force models may be switched on and off
- Foce model parameters read in from database
- ODE solver includes single-step (Runge-Kutta) and multi-step (Adams)

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RTGx Processing Flow

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- 1. SRIF Filter

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RTGx Processing Flow

- 1. SRIF Filter
- 2. UD Smoother with Ambiguity Resolution

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- 1. SRIF Filter
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- 3. SRIF Filter to apply ambiguity constraints

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- 1. SRIF Filter
- 2. UD Smoother with Ambiguity Resolution
- 3. SRIF Filter to apply ambiguity constraints
- 4. UD Smoother without Ambiguity Resolution

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RTGx Inputs

- 1. Parameters and options

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- 2. Deterministic state mapping functions, which come from integrating spacecraft state variational equations

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- 1. Parameters and options
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- 3. Partial derivatives of the measurements with respect to the problem parameters

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RTGx Inputs

- 1. Parameters and options
- 2. Deterministic state mapping functions, which come from integrating spacecraft state variational equations
- 3. Partial derivatives of the measurements with respect to the problem parameters
- 4. A time sequence of stochastic events for any of the stochastic parameters. As well as noting white noise and random walk updates, the time sequence will indicate parameters added and removed

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RTGx Optional Outputs

- 1. Estimates of linear combinations of state variables

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RTGx Optional Outputs

- 1. Estimates of linear combinations of state variables
- 2. Sigmas of linear combinations of state variables

GPS OCX

RTGx Optional Outputs

- 1. Estimates of linear combinations of state variables
- 2. Sigmas of linear combinations of state variables
- 3. SRI matrix

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RTGx Optional Outputs

- 1. Estimates of linear combinations of state variables
- 2. Sigmas of linear combinations of state variables
- 3. SRI matrix
- 4. Covariance matrix

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RTGx Optional Outputs

- 1. Estimates of linear combinations of state variables
- 2. Sigmas of linear combinations of state variables
- 3. SRI matrix
- 4. Covariance matrix
- 5. Sensitivity information

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RTGx Optional Outputs

- 1. Estimates of linear combinations of state variables
- 2. Sigmas of linear combinations of state variables
- 3. SRI matrix
- 4. Covariance matrix
- 5. Sensitivity information
- 6. Smoothing/forward smoothing/removal smoothing coefficients